

Sathya Kumaraguru

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Education

University of California, Santa Cruz | B.S. in Computer Science | GPA: 3.84

September 2024 – June 2026

Work Experience

Overture Maps Foundation | Project Researcher | Santa Cruz, CA

January 2025 – March 2025

- Designed and deployed a multi-signal ML system predicting operational status of 33M+ U.S. businesses at 90%+ accuracy by fusing geospatial features, satellite imagery (YOLO), and business metadata into a unified LightGBM inference engine.
- Built a full-stack geospatial platform in React + TypeScript with an integrated Python backend, surfacing real-time predictions on an H3 hexagonal grid with sub-second query performance across 33M+ records.
- Engineered scalable ETL pipelines joining multi-source datasets, eliminating preprocessing bottlenecks and cutting researcher turnaround time on new feature experiments.

Molina Healthcare | Software Engineer Intern | Remote

June 2024 – September 2024

- Built production automation tools in C#, ASP.NET, and SQL processing 500+ healthcare claims daily, replacing a fully manual workflow and reducing error rates in a live enterprise environment.
- Optimized SQL queries and stored procedures against large-scale datasets, improving throughput and cutting report generation time for operations and compliance teams.
- Partnered with senior engineers across Agile sprints to translate requirements into production-ready features, owning the full cycle from design doc through deployment.

Learning Support Services | Data Structures & Algorithms Tutor | Santa Cruz, CA

December 2025 – Present

- Tutor 20+ students weekly in DSA (C/C++), covering trees, graphs, dynamic programming, and complexity analysis — contributing to measurable improvements in course pass rates.
- Developed structured mental models for BFS/DFS, Dijkstra's, and DP recurrences, moving students from pattern memorization to independent problem-solving on timed assessments.

Projects

Paper | Privacy-first 3D asset optimizer — edge-compute pipeline, AI diagnostics, zero server round-trips

- Built a client-side WASM optimization pipeline (LOD swap → texture downsample → vertex quantization → re-analyze) and replaced global SSIM with windowed worst-decile weighting to catch localized mesh damage that traditional metrics miss.
- Integrated Llama 3.3 70B via Groq for structured AI asset diagnostics with Zod validation and a fault-tolerant 12-entry fallback cache — reflecting production-grade LLM integration patterns used across the industry.

Emplo | Agentic AI interview coach that adapts in real time to how each company actually hires

- Built a voice-driven interview simulator using ElevenLabs TTS with LLM-generated follow-up questions that adapt dynamically to user responses — applying agentic loop patterns powering next-gen AI assistants.
- Engineered a web scraper that grounds the LLM in live job posting data via RAG-style context injection, replacing generic prep with company-specific mock questions tailored to real hiring patterns.

MutationMap | Multimodal research tool fusing 3D spatial visualization with ML-based genomic classification

- Built an interactive 3D platform in React Three Fiber mapping multigene mutation patterns across chromosomal regions; implemented a Scikit-learn + Groq AI pipeline predicting cancer types at 92%+ accuracy, outperforming single-gene baselines.

Skills

- **Languages:** Python, C/C++, JavaScript, TypeScript, SQL
- **Frameworks & Tools:** PyTorch, Hugging Face, LangChain, LlamaIndex, FastAPI, React, PostgreSQL, Redis, Docker, AWS, Git